
Traffic Accident Severity Prediction

Using Machine Learning

Problem Id: 20

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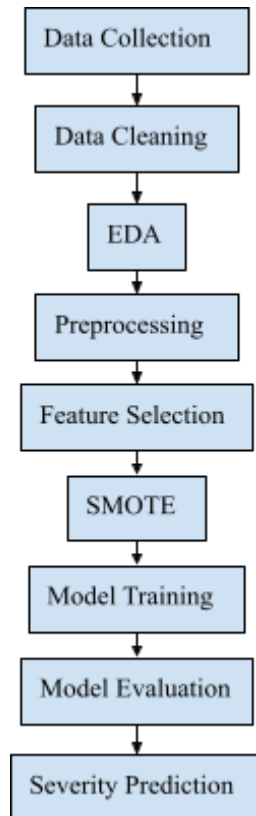
Problem Description

Traffic accidents are a major issue affecting road safety and public health. The severity of accidents depends on several factors such as vehicle speed, weather conditions, road type, and vehicle category. Predicting accident severity can help authorities and emergency services respond more efficiently and reduce damage and casualties.

This project aims to develop a machine learning system that predicts the severity of traffic accidents based on various input features. The system will use historical accident data and apply preprocessing techniques to clean the dataset and remove irrelevant records. Feature selection methods will be used to identify the most important factors affecting accident severity, such as speed and weather conditions.

Machine learning algorithms such as Logistic Regression and Support Vector Machine (SVM) will be used to build the prediction model. The SVM model will use different kernels such as Linear, Polynomial, and RBF to capture simple and complex patterns in the data. The performance of the model will be evaluated using metrics like confusion matrix and recall score to measure prediction accuracy.

Architecture



Dataset Collection

The road accident dataset was collected containing driver, vehicle, road, weather, and accident-related information.

Data Cleaning

Missing values, duplicate entries, and inconsistent categories were handled to improve data quality.

EDA (Exploratory Data Analysis)

The dataset was analyzed using graphs and visualizations to identify patterns, class imbalance, and important features.

Preprocessing

Categorical data was encoded and numerical data was scaled for machine learning model compatibility.

Feature Selection

Important features affecting accident severity were selected to reduce dimensionality and improve performance.

SMOTE

SMOTE was applied to balance the dataset by increasing minority class samples.

Model Training

Different SVM kernels (Linear, RBF, Polynomial, Sigmoid) and Logistic Regression were trained and tested.

Model Evaluation

Models were evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

Severity Prediction

The final system predicts whether an accident will result in slight or severe injury.

Models Used

1. Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning classification algorithm used to classify accident severity into slight or severe injury categories. SVM works by finding the optimal decision boundary (hyperplane) that separates data points of different classes. Different kernel functions such as Linear, RBF, Polynomial, and Sigmoid were used to analyze model performance on nonlinear and complex accident data patterns.

2. Logistic Regression

Logistic Regression is a supervised classification algorithm used for binary classification problems. It predicts the probability of an accident belonging to a particular severity class using a logistic function. Logistic Regression was used as a baseline model to compare performance with SVM models because of its simplicity, interpretability, and efficient classification capability.

Evaluation metrics used

1. Accuracy

Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted observations to the total observations.

$$Accuracy = \frac{TN + TP}{TP + TN + FP + FN}$$

Where

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

2. Precision

Precision measures how many predicted positive cases were actually positive.

$$Precision = \frac{TP}{TP + FP}$$

Higher precision indicates fewer false positive predictions.

3. Recall

Recall measures how many actual positive cases were correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

Higher recall indicates better detection of severe accident cases.

4. F1-Score

F1-Score is the harmonic mean of Precision and Recall. It provides a balanced measure of model performance.

$$F1_{score} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

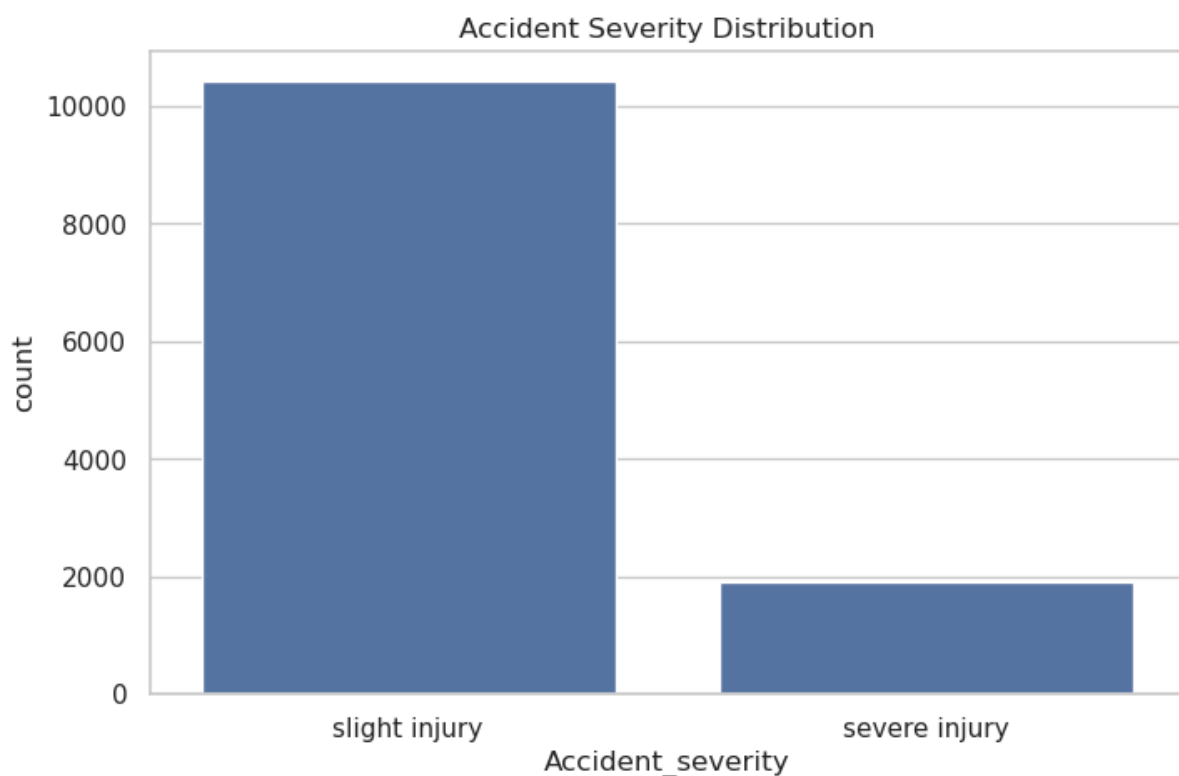
5. Confusion Matrix

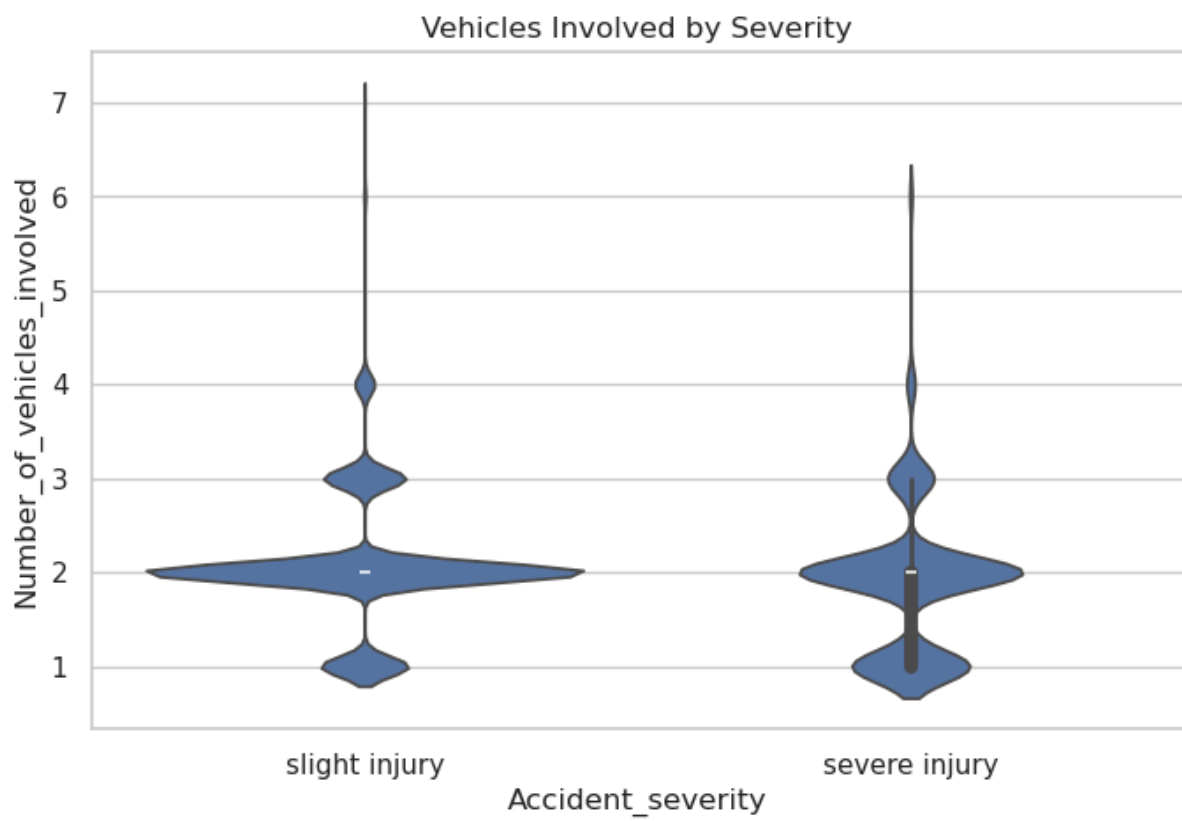
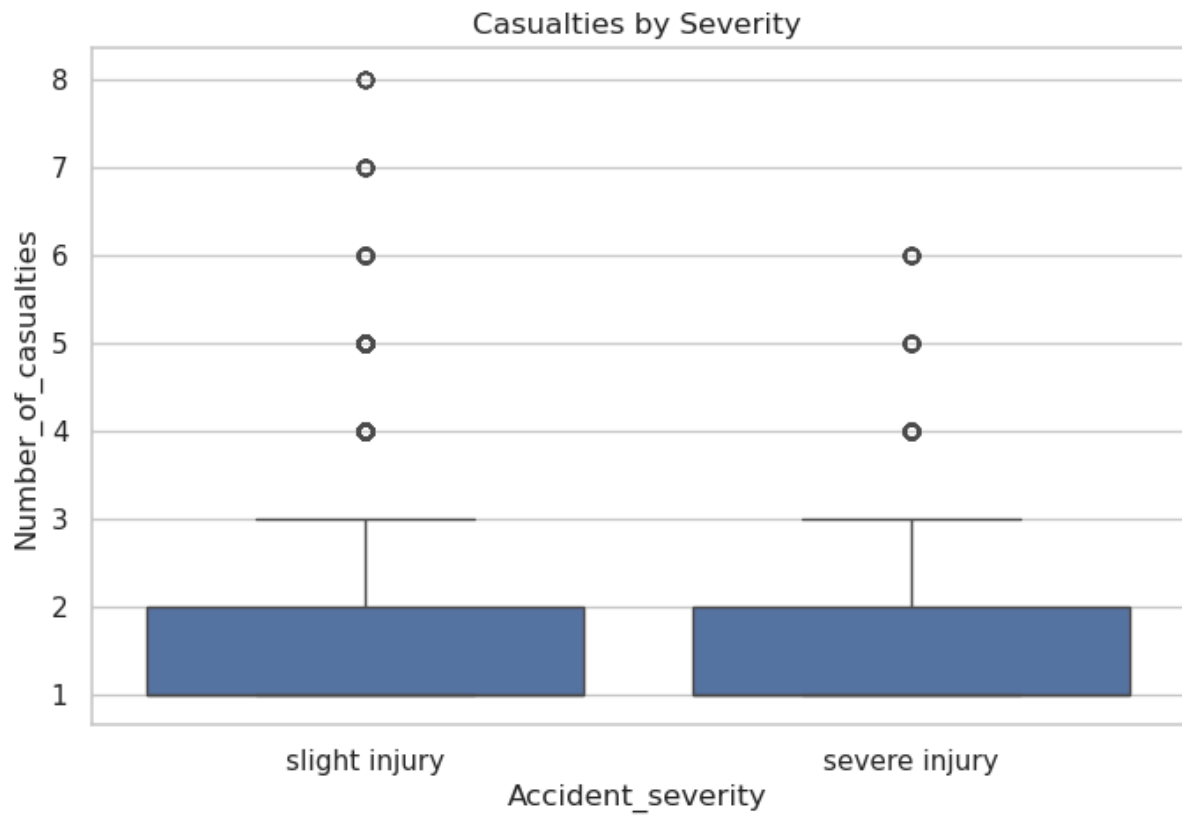
A Confusion Matrix is used to evaluate classification performance by showing the number of correct and incorrect predictions made by the model for each class. It helps analyze false positives and false negatives in accident severity prediction.

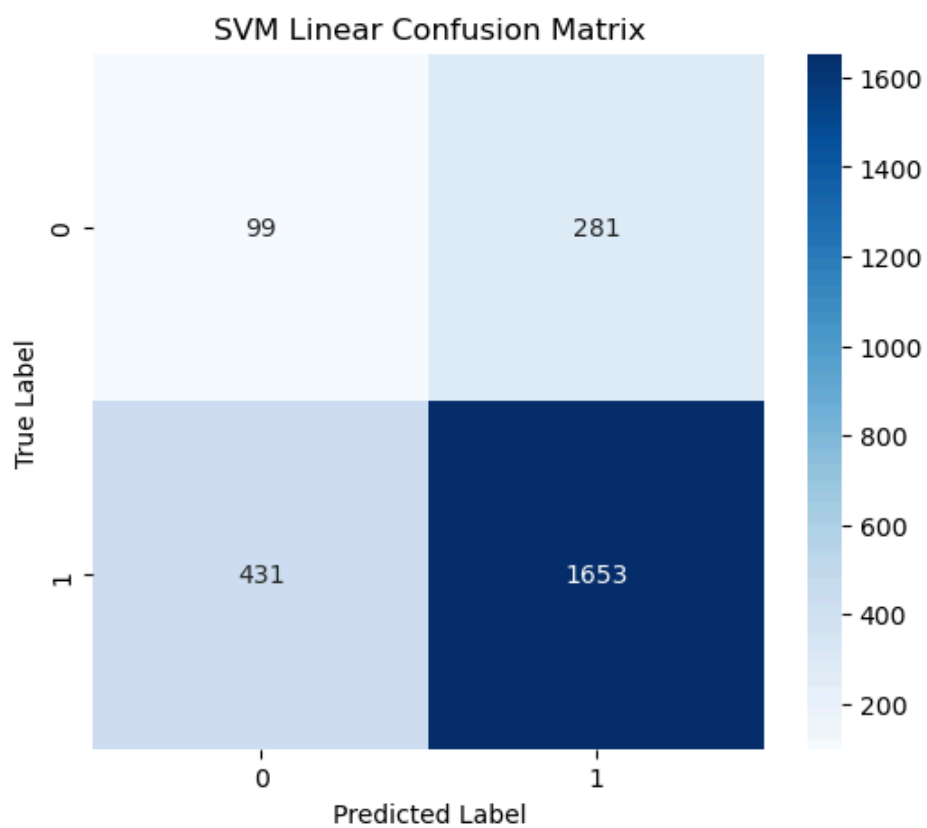
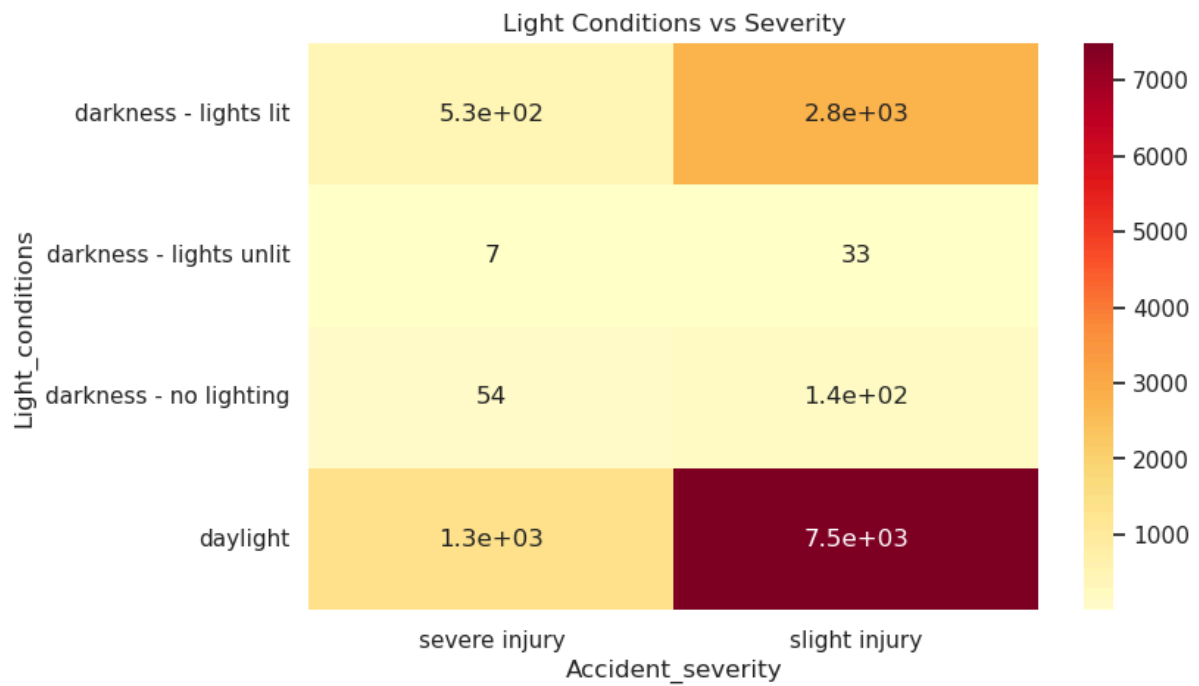
Preliminary Results and Visualizations

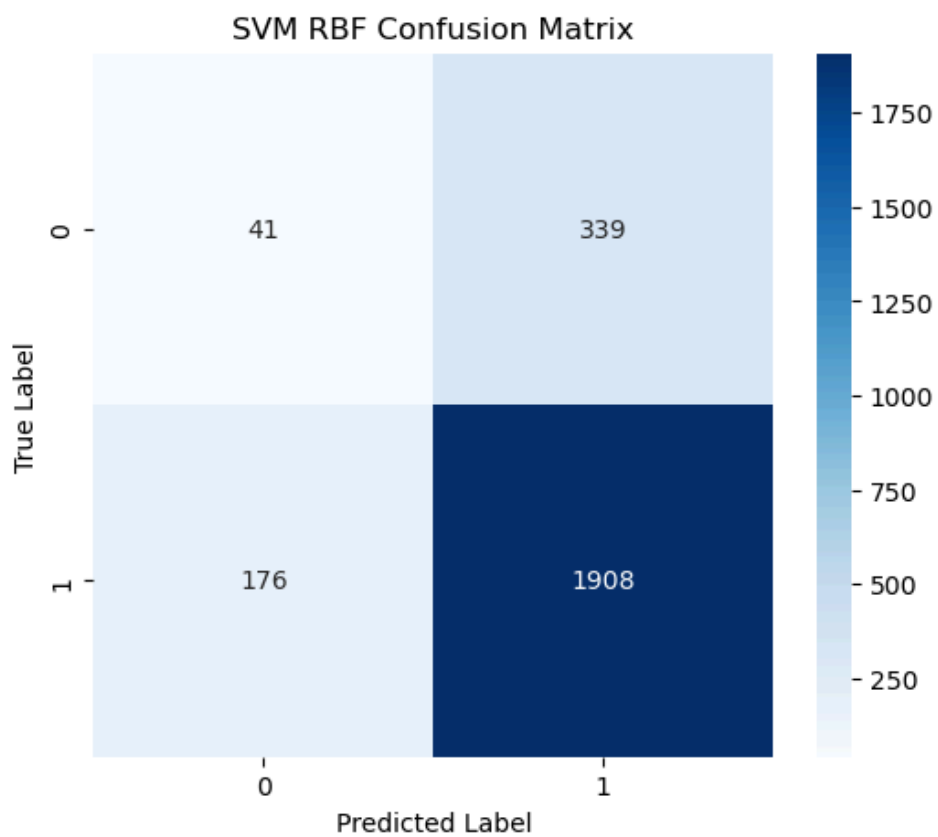
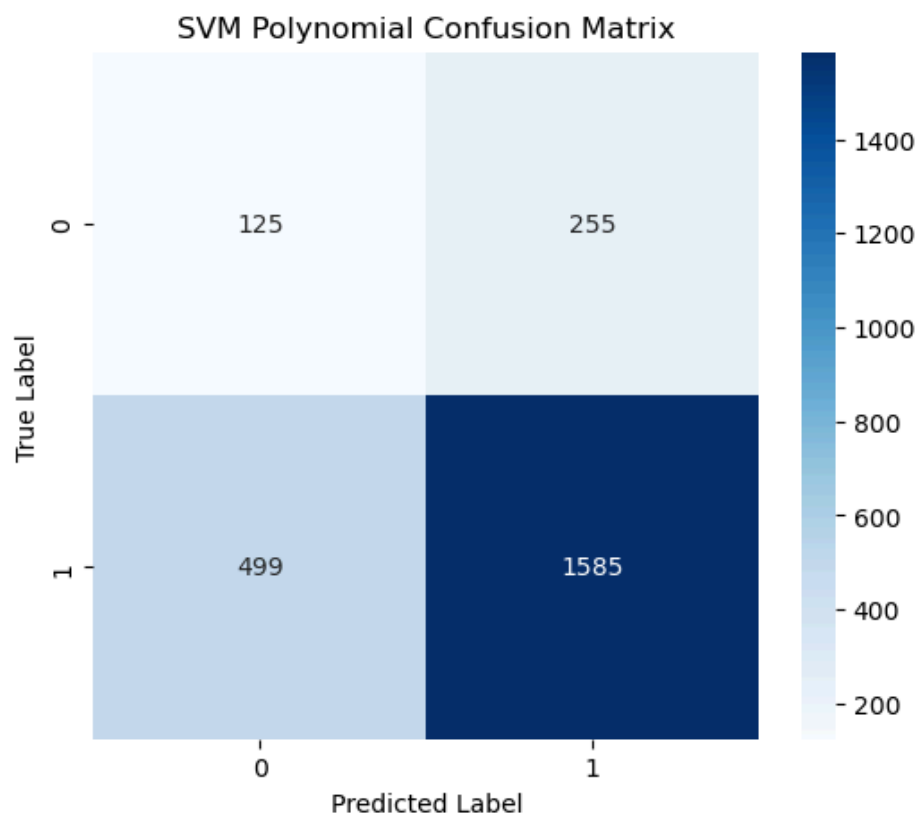
- The dataset showed a strong class imbalance where slight injury cases were significantly higher than severe injury cases.

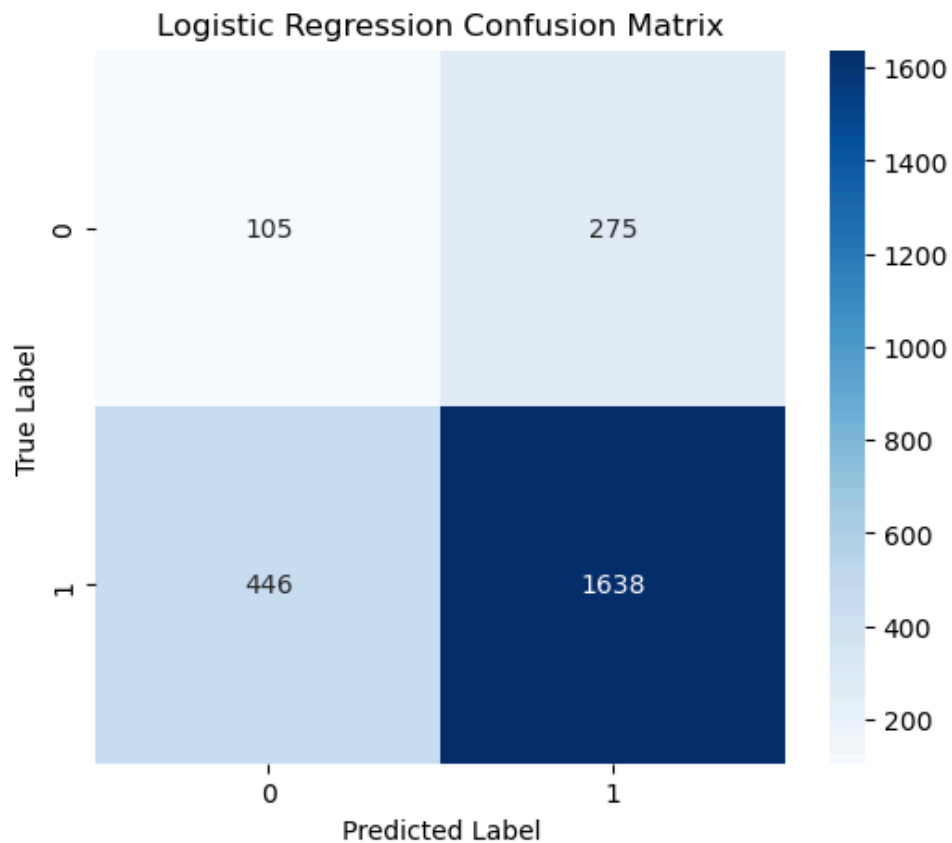
- Exploratory Data Analysis (EDA) was performed using bar plots, count plots, and heatmaps to understand feature distributions and relationships between accident-related factors.
- Data preprocessing techniques such as category reduction, one-hot encoding, feature selection, scaling, and SMOTE balancing improved the quality of training data.
- Multiple machine learning models including Support Vector Machine with Linear, RBF, Polynomial, and Sigmoid kernels and Logistic Regression were trained and evaluated.
- The RBF kernel achieved the highest overall accuracy, while the Polynomial kernel provided better severe injury detection and achieved the best macro-average F1-score.
- Confusion matrices and classification reports were used to compare model performance and analyze prediction behavior for slight and severe injury cases.











References

- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow – Aurélien Géron
- Pattern Recognition and Machine Learning – Christopher Bishop
- <https://scikit-learn.org>
- <https://www.kaggle.com>